

ICT703 Programming

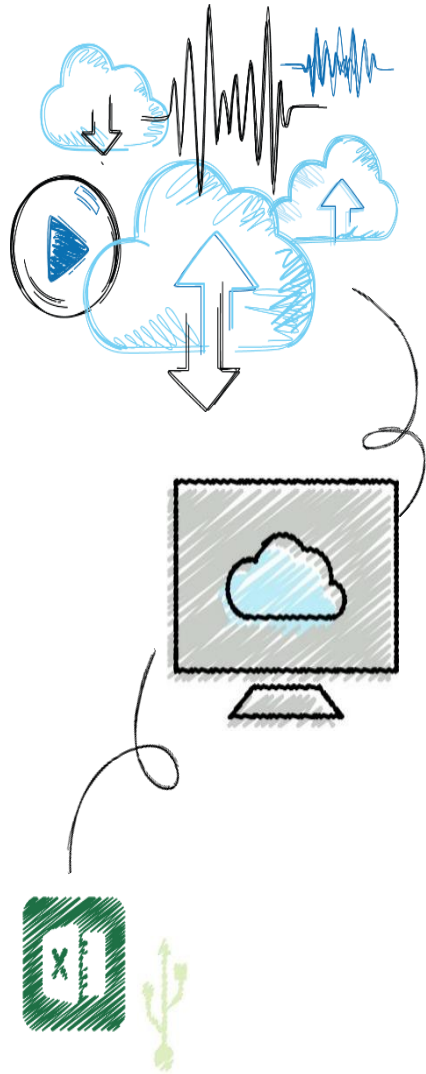
Module 3 Topic 3

Boolean Logic



Objectives (1 of 2)

- 1 Use logical operators to construct compound Boolean expressions





The Boolean Type, Comparisons, and Boolean Expressions

- **Boolean data type** consists of two values: true and false (typically through standard **True/False**)
- Example: **4 != 4** evaluates to **False**

Comparison Operator	Meaning
==	Equals
!=	Not equals
<	Less than
>	Greater than
<=	Less than or equal
>=	Greater than or equal



Logical Operators and Compound Boolean Expressions (1 of 4)

- Often a course of action must be taken if either of two conditions is true: Below are two approaches

```
number = int(input("Enter the numeric grade: "))
if number > 100:
    print("Error: grade must be between 100 and 0")
elif number < 0:
    print("Error: grade must be between 100 and 0")
else:
    # The code to compute and print the result goes here
```

- Simplified Code:

```
number = int(input("Enter the numeric grade: "))
if number > 100 or number < 0:
    print("Error: grade must be between 100 and 0")
else:
    # The code to compute and print the result goes here
```



Logical Operators and Compound Boolean Expressions (2 of 4)

A	B	A and B
True	True	True
True	False	False
False	True	False
False	False	False

Using AND:

Both have to be true to return true

A	B	A or B
True	True	True
True	False	True
False	True	True
False	False	False

Using OR:

Only one has to be true to return true

A	not A
True	False
False	True

Figure 3-5 The truth tables for **and**, **or**, and **not**



Logical Operators and Compound Boolean Expressions (3 of 4)

- Next example verifies some of the claims made in the previous truth tables:

A = True

B = False

print(A and B)

False

print(A or B)

True

print(not A)

False

- The logical operators are evaluated after comparisons but before the assignment operator

- not** has higher precedence than **and** and **or** (**not** will be executed first)

A = True

B = False

print(not A and B)

False

A = True

B = False

print(not (A and B))

True



Logical Operators and Compound Boolean Expressions (4 of 4)

Type of Operator	Operator Symbol
Exponentiation	**
Arithmetic negation	-
Multiplication, division, remainder	*, /, %
Addition, subtraction	+, -
Comparison	==, !=, <, >, <=, >=
Logical negation	not
Logical conjunction	and
Logical disjunction	or
Assignment	=



Short-Circuit Evaluation

- In **(A and B)**, if **A** is false, then so is the expression, and there is no need to evaluate **B**
 - In **(A or B)**, if **A** is true, then so is the expression, and there is no need to evaluate **B**
 - **Short-circuit evaluation:** Evaluation stops as soon as possible
 - Example:

```
count = int(input("Enter the count: "))
theSum = int(input("Enter the sum: "))
if count > 0 and theSum // count > 10:
    print("average > 10")
else:
    print("count = 0 or average <= 10")
```
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